

1) Solve each linear system using the method of substitution. Check your answers.

a) $\ell_1: y = 3x - 4$
 $\ell_2: x + y = 8$

$$\begin{aligned} \textcircled{1} y &= 3x - 4 \\ y &= 3(3) - 4 \\ y &= 5 \\ \textcircled{2} x + y &= 8 \\ x + (3x - 4) &= 8 \\ 4x - 4 &= 8 \\ 4x &= 12 \\ x &= 3 \end{aligned}$$

Solution: $x=3, y=5$

b) $\ell_1: x = -4y + 5$
 $\ell_2: x + 2y = 7$

$$\begin{aligned} \textcircled{1} x &= -4y + 5 \\ x &= -4(-1) + 5 \\ x &= 9 \\ \textcircled{2} x + 2y &= 7 \\ (-4y + 5) + 2y &= 7 \\ -2y + 5 &= 7 \\ -2y &= 2 \\ y &= -1 \end{aligned}$$

Solution: $x=9, y=-1$

c) $\ell_1: y = -2x + 3$
 $\ell_2: 4x - 3y = 1$

$$\begin{aligned} \textcircled{1} y &= -2x + 3 \\ y &= -2(1) + 3 \\ y &= 1 \\ \textcircled{2} 4x - 3y &= 1 \\ 4x - 3(-2x + 3) &= 1 \\ 4x + 6x - 9 &= 1 \\ 10x &= 10 \\ x &= 1 \end{aligned}$$

Solution: $x=1, y=1$

d) $\ell_1: 2x + 4y = 2$
 $\ell_2: x = 1 - 2y$

$$\begin{aligned} \textcircled{1} 2x + 4y &= 2 \\ 2(1 - 2y) + 4y &= 2 \\ 2 - 4y + 4y &= 2 \\ 0y &= 0 \end{aligned}$$

infinitely many solutions.
 the lines are parallel and coincident.

e) $\ell_1: x + 2y = 3$
 $\ell_2: 5x + 4y = 8$

$$\begin{aligned} \textcircled{1} x + 2y &= 3 \\ x &= 3 - 2y \\ x &= 3 - 2\left(\frac{7}{6}\right) \\ x &= 3 - \frac{7}{3} \\ x &= \frac{9}{3} - \frac{7}{3} \\ x &= \frac{2}{3} \\ \textcircled{2} 5x + 4y &= 8 \\ 5(3 - 2y) + 4y &= 8 \\ 15 - 10y + 4y &= 8 \\ -6y &= -7 \\ y &= \frac{7}{6} \end{aligned}$$

Solution: $x=\frac{2}{3}, y=\frac{7}{6}$

f) $\ell_1: 6x + 5y = 7$
 $\ell_2: x - y = 3$

$$\begin{aligned} \textcircled{1} 6x + 5y &= 7 \\ 6(3 + y) + 5y &= 7 \\ 18 + 6y + 5y &= 7 \\ 11y &= -11 \\ y &= -1 \\ \textcircled{2} x - y &= 3 \\ x &= 3 + y \\ x &= 3 + (-1) \\ x &= 2 \end{aligned}$$

Solution: $x=2, y=-1$

g) $\ell_1: 2m + n = 2$
 $\ell_2: 3m - 2n = 3$

h) $\ell_1: 3a + 2b = 4$
 $\ell_2: 2a + b = 6$

① $2m + n = 2$
 $n = 2 - 2m$
 $n = 2 - 2(1)$
 $n = 0$

② $3m - 2n = 3$
 $3m - 2(2 - 2m) = 3$
 $3m - 4 + 4m = 3$
 $7m = 7$
 $m = 1$

Solution: $m = 1, n = 0$

① $3a + 2b = 4$
 $3a + 2(6 - 2a) = 4$
 $3a + 12 - 4a = 4$
 $-a = -8$
 $a = 8$

② $2a + b = 6$
 $b = 6 - 2a$
 $b = 6 - 2(8)$
 $b = -10$

Solution: $a = 8, b = -10$

i) $\ell_1: 2x + y = 4$
 $\ell_2: 4x - y = 2$

① $2x + y = 4$
 $y = 4 - 2x$
 $y = 4 - 2(1)$
 $y = 2$

② $4x - y = 2$
 $4x - (4 - 2x) = 2$
 $4x - 4 + 2x = 2$
 $6x = 6$
 $x = 1$

Solution: $x = 1, y = 2$

2) Find the point of intersection of each pair of lines.

a) $\ell_1: 2x = y + 5$
 $\ell_2: 3x + y = -9$

b) $\ell_1: 4x + 2y = 7$
 $\ell_2: -x - 7y = 6$

① $2x = y + 5$
 $2x - 5 = y$
 $2(-\frac{4}{5}) - 5 = y$
 $-\frac{8}{5} - \frac{25}{5} = y$
 $-\frac{33}{5} = y$

② $3x + y = -9$
 $3x + (2x - 5) = -9$
 $5x - 5 = -9$
 $5x = -4$
 $x = -\frac{4}{5}$

Solution: $x = -\frac{4}{5}, y = -\frac{33}{5}$

① $4x + 2y = 7$
 $4(-7y - 6) + 2y = 7$
 $-28y - 24 + 2y = 7$
 $-26y = 31$
 $y = -\frac{31}{26}$

② $-x - 7y = 6$
 $-7y - 6 = x$
 $-7(-\frac{31}{26}) - 6 = x$
 $\frac{217}{26} - \frac{156}{26} = x$
 $x = \frac{61}{26}$

Solution: $x = \frac{61}{26}, y = -\frac{31}{26}$

c) $\ell_1: p + 4q = 3$
 $\ell_2: 5p = -2q + 3$

① $p + 4q = 3$
 $p = 3 - 4q$
 $p = 3 - 4\left(\frac{2}{3}\right)$
 $p = \frac{9}{3} - \frac{8}{3}$
 $p = \frac{1}{3}$

② $5p = -2q + 3$
 $5(3 - 4q) = -2q + 3$
 $15 - 20q = -2q + 3$
 $12 = 18q$
 $\frac{12}{18} = q$
 $q = \frac{2}{3}$

solution: $p = \frac{1}{3}, \frac{2}{3}$

d) $\ell_1: a + b + 6 = 0$
 $\ell_2: 2a - b - 3 = 0$

① $a + b + 6 = 0$
 $a + (2a - 3) + 6 = 0$
 $3a + 3 = 0$
 $3a = -3$
 $a = -1$

② $2a - b - 3 = 0$
 $2a - 3 = b$
 $2(-1) - 3 = b$
 $b = -5$

solution: $a = -1, b = -5$

e) $\ell_1: x - 2y - 2 = 0$
 $\ell_2: 3x + 4y - 16 = 0$

① $x - 2y - 2 = 0$
 $x = 2y + 2$
 $x = 2(1) + 2$
 $x = 4$

② $3x + 4y - 16 = 0$
 $3(2y + 2) + 4y - 16 = 0$
 $6y + 6 + 4y - 16 = 0$
 $10y - 10 = 0$
 $10y = 10$
 $y = 1$

solution: $x = 4, y = 1$

3) Samantha works twice as many hours per week as Adriana. Together they work a total of 39 hours in one week.

a) Write an equation to represent the information in the first sentence.

$S = 2A$

b) Write an equation to represent the information in the second sentence.

$S + A = 39$

c) Use the method of substitution to find the number of hours worked by each person.

$$\begin{array}{ll} S = 2A & S + A = 39 \\ S = 2(13) & 2A + A = 39 \\ S = 26 & 3A = 39 \\ & A = 13 \end{array}$$

Solution: Samantha worked 26 hours
Adriana worked 13 hours

4) Ugo plays hockey and is awarded 2 points for each goal and 1 point for each assist. Last season he had a total of 86 points. He scored 17 fewer goals than assists.

a) Write a linear system to represent the information

$$\begin{array}{l} \textcircled{1} \quad 2g + a = 86 \\ \textcircled{2} \quad a - g = 17 \end{array}$$

b) Solve the system

$$\begin{array}{ll} 2g + a = 86 & a - g = 17 \\ 2g + (17 + g) = 86 & a = 17 + g \\ 3g = 69 & a = 17 + 23 \\ g = 23 & a = 40 \end{array}$$

c) What does the solution represent in the context of this question?

Ugo scored 23 goals and had 40 assists.

5) Joanne's family decides to rent a hall for her retirement party. Pin Hall charges \$500 for the hall and \$15 per meal. Bloom Place charges \$350 for the hall and \$18 per meal.

a) Write two equations to represent the information.

$$\textcircled{1} \quad C = 500 + 15g$$

$$\textcircled{2} \quad C = 350 + 18g$$

b) Solve the linear system to find the number of guests for which the charges are the same at both halls.

$$\begin{aligned} C &= 500 + 15g \\ 350 + 18g &= 500 + 15g \\ 3g &= 150 \\ g &= 50 \end{aligned}$$

$$\begin{aligned} C &= 350 + 18g \\ C &= 350 + 18(50) \\ C &= 1250 \end{aligned}$$

Solution: Both halls would charge \$1250 if 50 guests attended.

Answers:

1) a) $x = 3, y = 5$ b) $x = 9, y = -1$ c) $x = 1, y = 1$ d) infinite solutions

e) $x = \frac{2}{3}, y = \frac{7}{6}$ f) $x = 2, y = -1$ g) $m = 1, n = 0$ h) $a = 8, b = -10$ i) $x = 1, y = 2$

2) a) $x = -\frac{4}{5}, y = -\frac{33}{5}$ b) $x = \frac{61}{26}, y = -\frac{31}{26}$ c) $p = \frac{1}{3}, q = \frac{2}{3}$ d) $a = -1, b = -5$ e) $x = 4, y = 1$

3) a) $S = 2a$ b) $S + A = 39$ c) Samantha worked 26 hours and Adriana worked 13 hours

4) a) $2g + a = 86; g = a - 17$ b) $g = 23, a = 40$ c) 23 goals; 40 assists

5) a) $C = 500 + 15n; C = 350 + 18n$ b) 50 guests